# The gyroscopic matrix $\mathbf{N}$: how the fluid core and damping enter

Working notes (prepared with Claude) — to be checked

1 October 2026

This note expands on the matrix $\mathbf{N}$ of §10.1 of `rotation_dynamic_elastic.pdf` (equation numbers prefixed by "N" refer to that note). $\mathbf{N}$ multiplies $\dot{\boldsymbol{\omega}}$ in the rotational row of the weak form,

$$(\tilde{\mathbf{I}}\boldsymbol{\omega} + \mathbf{N}\dot{\boldsymbol{\omega}} - \mathbf{j} + \mathbf{k}) \cdot \boldsymbol{\omega}' = 0, \tag{1}$$

and carries everything about the inertia of the rotation vector itself. The fluid core and the damping each change $\mathbf{N}$, in different ways: the core rescales it, and damping adds a new part. Throughout, $\mathsf{E} = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, with $\mathsf{E}^2 = -1$, $\mathsf{E}^T = -\mathsf{E}$ and $\mathsf{E}^{-1} = -\mathsf{E}$. The equatorial (top-left $2 \times 2$) block of $\mathbf{N}$ is written $\mathbf{N}_\perp$.

## 1 Where N comes from: no core, no damping

1. Start from the equatorial Liouville equation, eq. N(10), with $\mathbf{h} = \boldsymbol{\Gamma} = \mathbf{0}$:

$$I_1\dot{\boldsymbol{\omega}} + \dot{\mathbf{j}} = \Omega\mathsf{E}\big[(I_3 - I_1)\boldsymbol{\omega} - \mathbf{j}\big]. \tag{2}$$

2. Drop $\dot{\mathbf{j}}$, which is $O(\nu/\Omega)$ relative to the retained terms (N§4.2):

$$I_1\dot{\boldsymbol{\omega}} = \Omega\mathsf{E}\big[(I_3 - I_1)\boldsymbol{\omega} - \mathbf{j}\big]. \tag{3}$$

3. Multiply both sides on the left by $\mathsf{E}^{-1}/\Omega = -\mathsf{E}/\Omega$. On the right this removes $\Omega\mathsf{E}$:

$$-\frac{I_1}{\Omega}\mathsf{E}\dot{\boldsymbol{\omega}} = (I_3 - I_1)\boldsymbol{\omega} - \mathbf{j}. \tag{4}$$

4. Add $(I_1/\Omega)\mathsf{E}\dot{\boldsymbol{\omega}} + \mathbf{j}$ to both sides:

$$\boxed{(I_3 - I_1)\boldsymbol{\omega} + \frac{I_1}{\Omega}\mathsf{E}\,\dot{\boldsymbol{\omega}} = \mathbf{j}} \tag{5}$$

5. Comparing with eq. (1) (with $\mathbf{k} = \mathbf{0}$), the equatorial block of $\tilde{\mathbf{I}}$ is $(I_3 - I_1)1$ and

$$\mathbf{N}_\perp = \frac{I_1}{\Omega}\,\mathsf{E}. \tag{6}$$

   This is a multiple of $\mathsf{E}$, and therefore antisymmetric.

6. **Axial part.** The third row and column of $\mathbf{N}$ are zero, so the axial component satisfies $-I_3\omega_3 = j_3$. This is algebraic: $\omega_3$ has no memory.

7. **Static limit.** Setting $\dot{\boldsymbol{\omega}} = \mathbf{0}$ recovers AA24's static relation $\tilde{\mathbf{I}}\boldsymbol{\omega} = \mathbf{j}$.

## 2 The fluid core: it rescales N

1. With the core, the relative angular momentum is no longer zero. To leading order in $\nu/\Omega$ (N§4.1),
$$\mathbf{h}_{\rm core} \approx \frac{A_f}{\Omega}\,\mathsf{E}\dot{\boldsymbol{\omega}}, \tag{7}$$
where $A_f$ is the fluid core's equatorial moment. The core's rotation differs from the mantle's by only $O(\epsilon)$ of $\boldsymbol{\omega}$.

2. This is small, but it enters the Liouville equation, eq. N(10), through the term $-\Omega\mathsf{E}\mathbf{h}$:
$$I_1\dot{\boldsymbol{\omega}} + \dot{\mathbf{h}} = \Omega\mathsf{E}\big[(I_3 - I_1)\boldsymbol{\omega} - \mathbf{j}\big] - \Omega\mathsf{E}\mathbf{h}. \tag{8}$$

3. Evaluate the new term. The factor $\Omega$ cancels the $1/\Omega$, and $\mathsf{E}^2 = -1$ flips the sign:
$$-\Omega\mathsf{E}\mathbf{h}_{\rm core} = -\Omega\mathsf{E}\,\frac{A_f}{\Omega}\mathsf{E}\dot{\boldsymbol{\omega}} = -A_f\mathsf{E}^2\dot{\boldsymbol{\omega}} = +A_f\dot{\boldsymbol{\omega}}. \tag{9}$$

4. Move it to the left-hand side:
$$(I_1 - A_f)\dot{\boldsymbol{\omega}} + \dot{\mathbf{h}}_{\rm core} = \Omega\mathsf{E}\big[(I_3 - I_1)\boldsymbol{\omega} - \mathbf{j}\big]. \tag{10}$$

5. $\dot{\mathbf{h}}_{\rm core} = (A_f/\Omega)\mathsf{E}\ddot{\boldsymbol{\omega}}$ is a further factor $\nu/\Omega$ smaller than $A_f\dot{\boldsymbol{\omega}}$, so drop it. With $A_m := I_1 - A_f$, the mantle's equatorial moment,
$$A_m\dot{\boldsymbol{\omega}} = \Omega\mathsf{E}\big[(I_3 - I_1)\boldsymbol{\omega} - \mathbf{j}\big]. \tag{11}$$

6. Repeating steps 3–5 of §1 with $I_1$ replaced by $A_m$:
$$\boxed{\mathbf{N}_\perp = \frac{A_m}{\Omega}\,\mathsf{E}} \tag{12}$$

7. **Interpretation.** The core only rescales $\mathbf{N}$. It stays a multiple of $\mathsf{E}$, and so stays antisymmetric. Writing $A_{\rm w}$ for "$I_1$ or $A_m$", and allowing a triaxial Earth, the full matrix is
$$\mathbf{N} = \frac{1}{\Omega}\begin{pmatrix} 0 & -A_{w,2} & 0 \\ A_{w,1} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \tag{13}$$

8. Nothing else in the problem changes. The static limit is untouched, since it does not involve $\mathbf{N}$, and the Euler frequency becomes $\sigma_{\rm e} = \Omega(I_3 - I_1)/A_{\rm w}$. PREM gives $A_f/I_1 = 0.113$.

## 3 Damping: it adds a symmetric part to N

1. The solver uses the damped law (N§8.1)
$$\dot{\boldsymbol{\omega}} = \mathsf{L}(\boldsymbol{\omega} - \mathbf{q}), \qquad \mathsf{L} = \sigma_{\rm e}\mathsf{E} - \gamma 1, \qquad \mathbf{q} = \frac{\mathbf{j}}{I_3 - I_1}. \tag{14}$$
To put it in the form of eq. (1), invert $\mathsf{L}$.

2. Multiply $\mathsf{L}$ by $-\sigma_{\rm e}\mathsf{E} - \gamma 1$, using $\mathsf{E}^2 = -1$:
$$(\sigma_{\rm e}\mathsf{E} - \gamma 1)(-\sigma_{\rm e}\mathsf{E} - \gamma 1) = -\sigma_{\rm e}^2\mathsf{E}^2 - \sigma_{\rm e}\gamma\mathsf{E} + \sigma_{\rm e}\gamma\mathsf{E} + \gamma^2 1 = (\sigma_{\rm e}^2 + \gamma^2)1. \tag{15}$$

3. Hence
$$\mathsf{L}^{-1} = -\frac{\sigma_{\rm e}\mathsf{E} + \gamma 1}{\sigma_{\rm e}^2 + \gamma^2}. \tag{16}$$

4. Multiply the damped law on the left by $\mathsf{L}^{-1}$: $\mathsf{L}^{-1}\dot{\boldsymbol{\omega}} = \boldsymbol{\omega} - \mathbf{q}$.

5. Multiply by $-(I_3 - I_1)$ and use $(I_3 - I_1)\mathbf{q} = \mathbf{j}$:
$$\boxed{(I_3 - I_1)\boldsymbol{\omega} + \frac{(I_3 - I_1)(\sigma_{\rm e}\mathsf{E} + \gamma 1)}{\sigma_{\rm e}^2 + \gamma^2}\,\dot{\boldsymbol{\omega}} = \mathbf{j}} \tag{17}$$
So with damping,
$$\mathbf{N}_\perp = \frac{(I_3 - I_1)(\sigma_{\rm e}\mathsf{E} + \gamma 1)}{\sigma_{\rm e}^2 + \gamma^2}. \tag{18}$$

6. **Check.** At $\gamma = 0$: $\mathbf{N}_\perp = (I_3 - I_1)\mathsf{E}/\sigma_{\rm e} = (I_3 - I_1)\mathsf{E}\,A_{\rm w}/[\Omega(I_3 - I_1)] = (A_{\rm w}/\Omega)\mathsf{E}$, as in §§1–2.

7. **Split into two parts.** $\mathbf{N}_\perp = \mathbf{N}_a + \mathbf{N}_s$, with
$$\mathbf{N}_a = \frac{(I_3 - I_1)\sigma_{\rm e}}{\sigma_{\rm e}^2 + \gamma^2}\,\mathsf{E} = \frac{A_{\rm w}}{\Omega}\,\frac{1}{1 + \gamma^2/\sigma_{\rm e}^2}\,\mathsf{E} \qquad \text{(antisymmetric, gyroscopic)}, \tag{19}$$
$$\mathbf{N}_s = \frac{(I_3 - I_1)\gamma}{\sigma_{\rm e}^2 + \gamma^2}\,1 \qquad \text{(symmetric, dissipative)}. \tag{20}$$

8. **Sizes.** With $\gamma = \sigma_{\rm e}/(2Q)$, $\gamma/\sigma_{\rm e} = 1/(2Q) = 0.005$ for $Q = 100$.

   - $\mathbf{N}_a$ is the undamped $\mathbf{N}$ reduced by the factor $1/(1 + \gamma^2/\sigma_{\rm e}^2)$, i.e. by $2.5 \times 10^{-5}$. This is negligible.
   - $\mathbf{N}_s$ is new. Its size relative to $\mathbf{N}_a$ is $\gamma/\sigma_{\rm e} = 1/(2Q)$.

# 4 Why the split matters: an energy identity

1. Consider free motion: no direct load, so $\mathbf{j} = (I_3 - I_1)\mathsf{K}\boldsymbol{\omega}$ (N§6.2). The equatorial rotational row becomes
$$(I_3 - I_1)(1 - \mathsf{K})\boldsymbol{\omega} + \mathbf{N}\dot{\boldsymbol{\omega}} = \mathbf{0}. \tag{21}$$

2. Take the dot product with $\dot{\boldsymbol{\omega}}$:
$$(I_3 - I_1)\,\dot{\boldsymbol{\omega}}^T(1 - \mathsf{K})\boldsymbol{\omega} + \dot{\boldsymbol{\omega}}^T\mathbf{N}_a\dot{\boldsymbol{\omega}} + \dot{\boldsymbol{\omega}}^T\mathbf{N}_s\dot{\boldsymbol{\omega}} = 0. \tag{22}$$

3. The antisymmetric part does no work. $\dot{\boldsymbol{\omega}}^T\mathbf{N}_a\dot{\boldsymbol{\omega}}$ is a number, so it equals its transpose, $\dot{\boldsymbol{\omega}}^T\mathbf{N}_a^T\dot{\boldsymbol{\omega}} = -\dot{\boldsymbol{\omega}}^T\mathbf{N}_a\dot{\boldsymbol{\omega}}$. It is therefore zero.

4. $1 - \mathsf{K}$ is symmetric, so $\dot{\boldsymbol{\omega}}^T(1 - \mathsf{K})\boldsymbol{\omega} = \frac{\rm d}{{\rm d}t}\left[\frac{1}{2}\boldsymbol{\omega}^T(1 - \mathsf{K})\boldsymbol{\omega}\right]$.

5. Hence
$$\boxed{\frac{\rm d}{{\rm d}t}\left[\tfrac{1}{2}(I_3 - I_1)\,\boldsymbol{\omega}^T(1 - \mathsf{K})\boldsymbol{\omega}\right] = -\dot{\boldsymbol{\omega}}^T\mathbf{N}_s\dot{\boldsymbol{\omega}} \le 0} \tag{23}$$
The inequality holds because $\mathbf{N}_s$ is a positive multiple of 1 when $\gamma > 0$.

6. **Consequences.**

   - With $\gamma = 0$ the bracketed "energy" is conserved. This is the conserved quantity that makes the free orbit an ellipse in the detailed derivation note of §7.
   - With $\gamma > 0$, $\mathbf{N}_s$ drains it, so the wobble decays.
   - The core changes only $\mathbf{N}_a$. It therefore changes the wobble frequency but not the decay.

## 5 Why the main note writes things in terms of N

1. The reciprocity argument (N§10.2) needs only $\mathbf{N}$. The adjoint problem uses $\mathbf{N}^\dagger = -\mathbf{N}^T$.

2. Transposing flips the sign of the antisymmetric part and keeps the symmetric part, so

$$-\mathbf{N}^T = -(\mathbf{N}_a + \mathbf{N}_s)^T = \mathbf{N}_a - \mathbf{N}_s. \tag{24}$$

3. Run backwards in time, $\mathrm{d}/\,\mathrm{d}t \to -\,\mathrm{d}/\,\mathrm{d}\tau$, this becomes $-\mathbf{N}_a + \mathbf{N}_s$ (N§10.3):

   - the gyroscopic part is reversed, i.e. the rotation is reversed;
   - the dissipative part is kept, so the adjoint is damped in reversed time just as the forward problem is in forward time.

4. Both the core (a rescaled $\mathbf{N}_a$) and the damping (an added $\mathbf{N}_s$) therefore fit the same reciprocity theorem, with no new derivation needed.

5. In the forward solver itself, the equivalent form $\dot{\boldsymbol{\omega}} = \mathsf{L}(\boldsymbol{\omega} - \mathbf{q})$, with $\sigma_\mathrm{e}$ built from $A_\mathrm{w}$, is simpler to use (N§8). The two forms describe the same equation.